

Python Variables

Bit by Bit Coding - Term 2 | Taster 1

Key Concepts

- `print()` displays text/numbers on the screen.
- A variable is a named box that stores data: `name = value`.
- Data types: `str` (text), `int` (whole numbers), `float` (decimals), `bool` (`True/False`).
- `input()` reads what the user types - it ALWAYS returns a string.
- `type()` tells you the data type of any value.
- f-strings (`f"..."`) let you drop variables straight into text.

Syntax Reference

Output & Input

```
print("Hello World")
name = input("Enter your name: ")
age = int(input("Enter your age: ")) # must convert!
print(type(age))                    # <class 'int'>
```

Variables & Types

```
score = 95           # int
price = 19.99         # float
word = "Python"       # str
is_ready = True       # bool
print(type(price))    # <class 'float'>
```

Arithmetic Operators

+ add, - subtract, * multiply, / divide, // floor divide, % remainder, ** power.

```
print(7 + 2)         # 9
print(7 / 2)         # 3.5
print(7 // 2)        # 3   (floor division)
print(7 % 2)         # 1   (remainder)
print(2 ** 3)        # 8   (2 to the power 3)
```

f-strings

```
name = "Aisha"
score = 87
print(f"{name} scored {score} points.")
print(f"Half price: ${19.99 / 2:.2f}") # $10.00
```

Common Patterns

- Update a variable: `score = score + 10` (or shorthand `score += 10`).
- Convert input to a number before doing maths: `int(input(...))`.
- Use f-strings to build clear, readable messages.

Tips & Common Mistakes

- `"10" + "5"` gives `"105"` (text joining), but `10 + 5` gives `15` (maths). Watch types!
- `=` assigns a value; `==` compares two values. They are different.
- Variable names: use `snake_case`, can't start with a number, and are case-sensitive.

- `input()` returns a string even when the user types a number - convert it first.